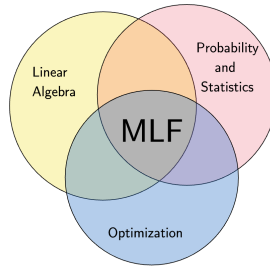


Week-4, Session-1



Karthik Thiagarajan

Disclaimer: These notes were generated as a part of this [live session](#). It can't be used as a standalone resource or a set of notes for this week.

1. Nullspace correction (last session)

$$A \rightarrow m \times n$$

$$\mathcal{N}(A) = \{x : Ax = 0, x \in \mathbb{R}^n\}$$

When is $\mathcal{N}(A) = \{0\}$?

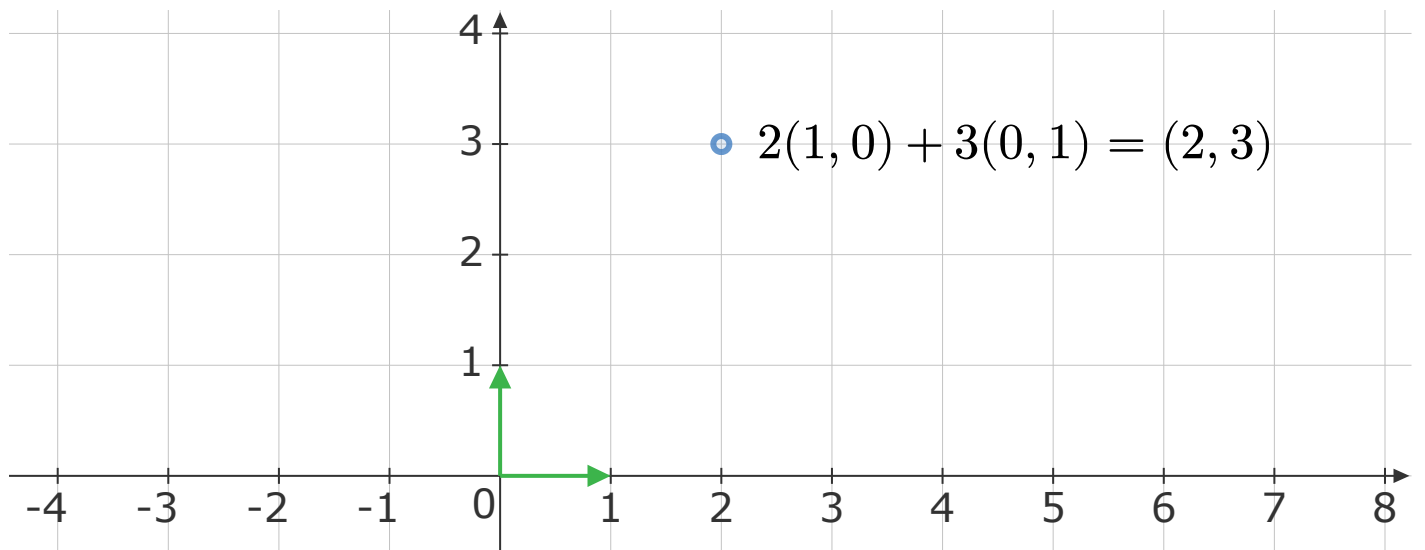
When the columns of A are linearly independent.

At 18:00 minutes in the previous session.

2. Linear Transformations

Every linear transformation (linear map) can be thought of as a matrix. In this case, we are interested in linear maps that go from $\mathbb{R}^n$ to $\mathbb{R}^n$.

Recall that a linear map can be exactly determined by what it does to the basis vectors in the domain. This is because every vector can be written as a unique linear combination of the basis vectors.

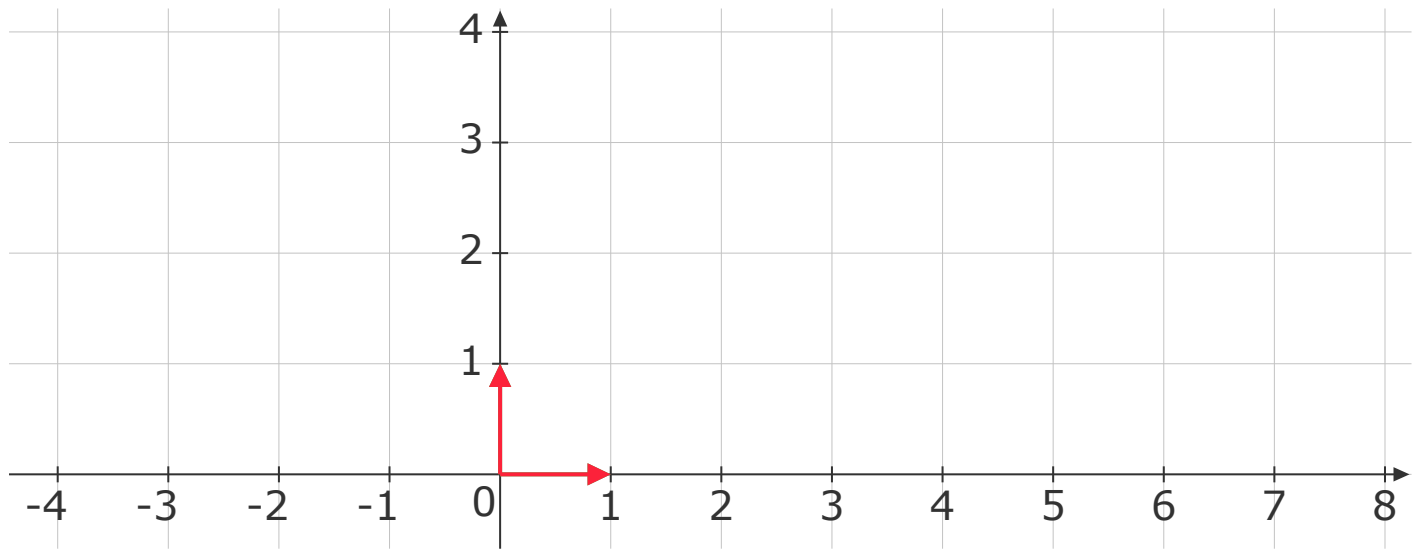


The columns of the matrix tell you where the basis vectors are going to go after the transformation.

2.1. Example-1

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

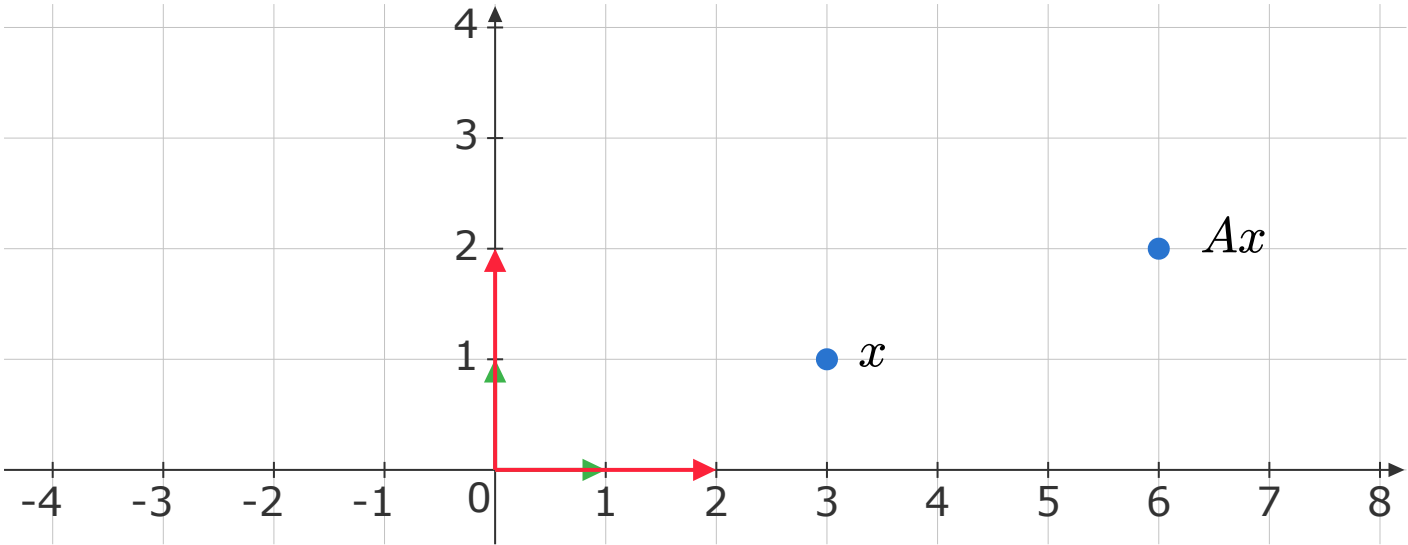
A is the identity map or the identity matrix.



2.2. Example-2

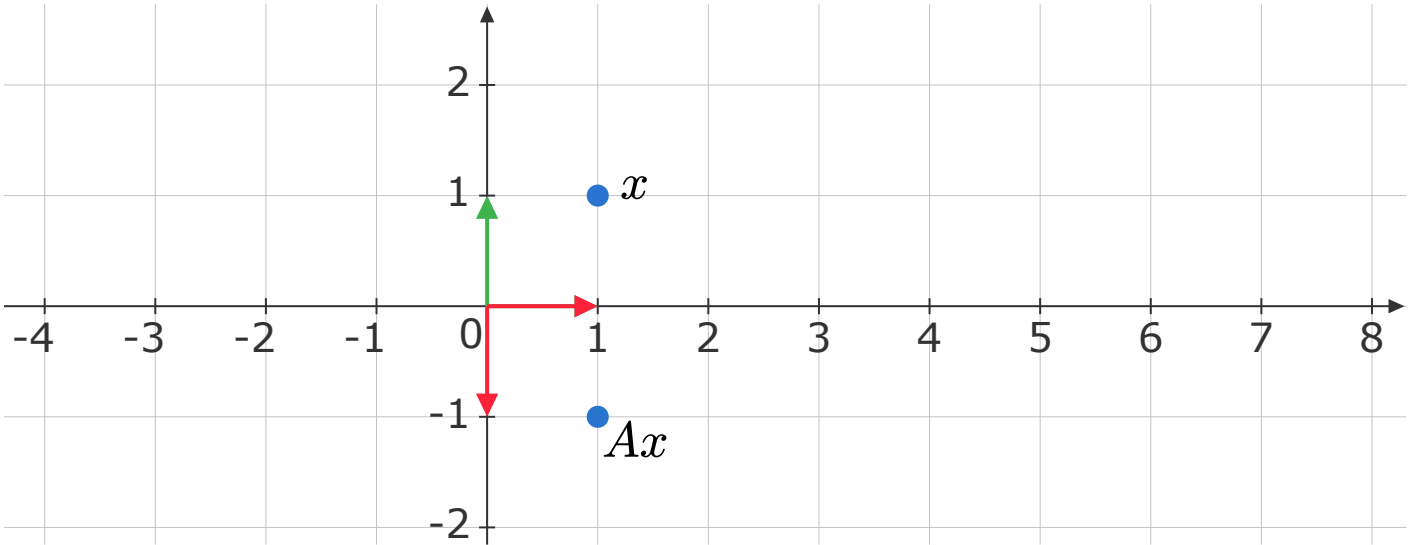
$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$$

A is scaling the basis vectors.



2.3. Example-3

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

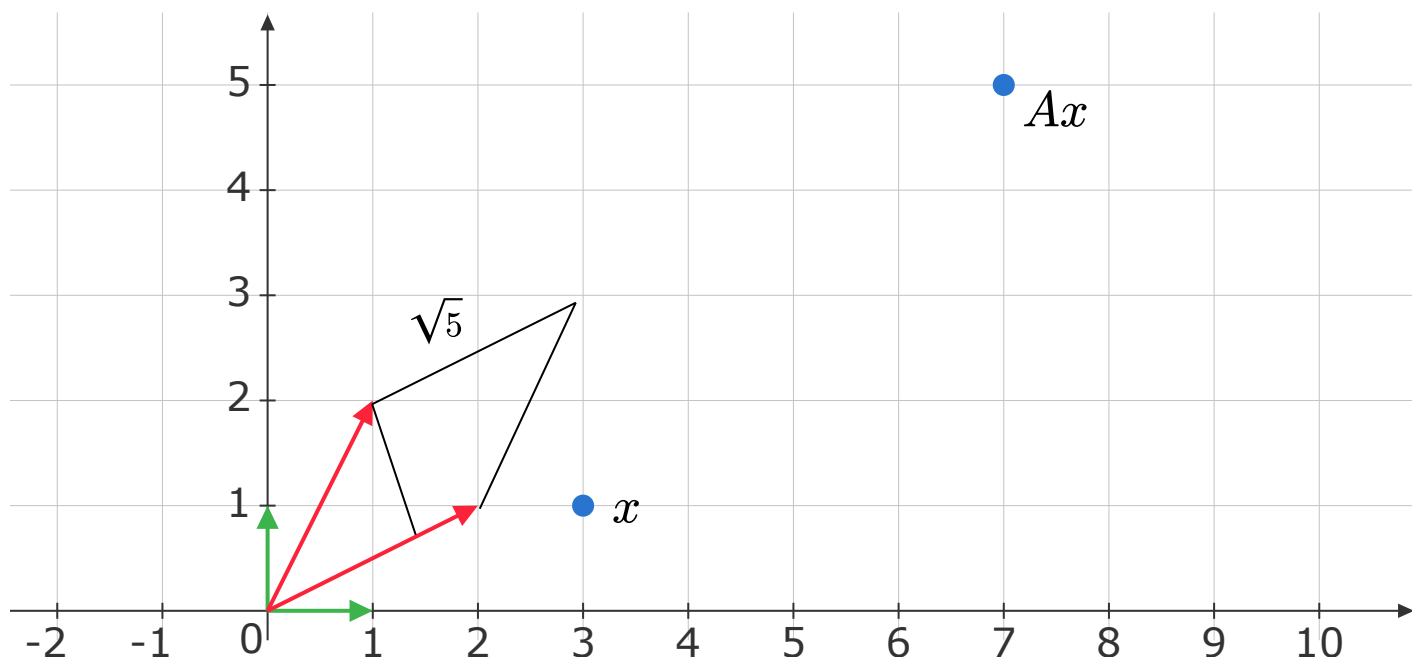


What is A doing to the basis vectors here? Rotation clockwise by 90° .

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

2.4. Example-4

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$



$$A \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$$

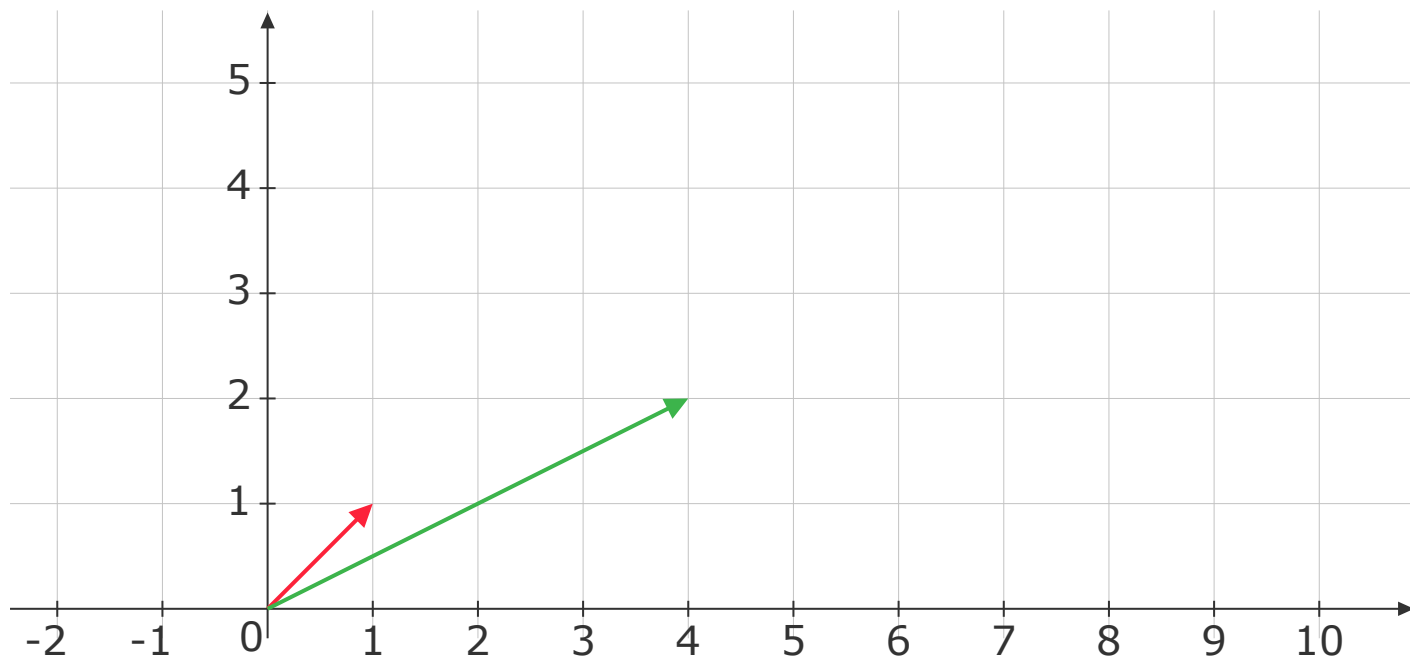
You can use this to understand determinants geometrically.

$$|A| = 3$$

The signed area of the parallelogram is actually equal to $|A|$. For a matrix that is not invertible, the area will turn out to be 0.

3. Eigenvalues and eigenvectors: Motivation

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$



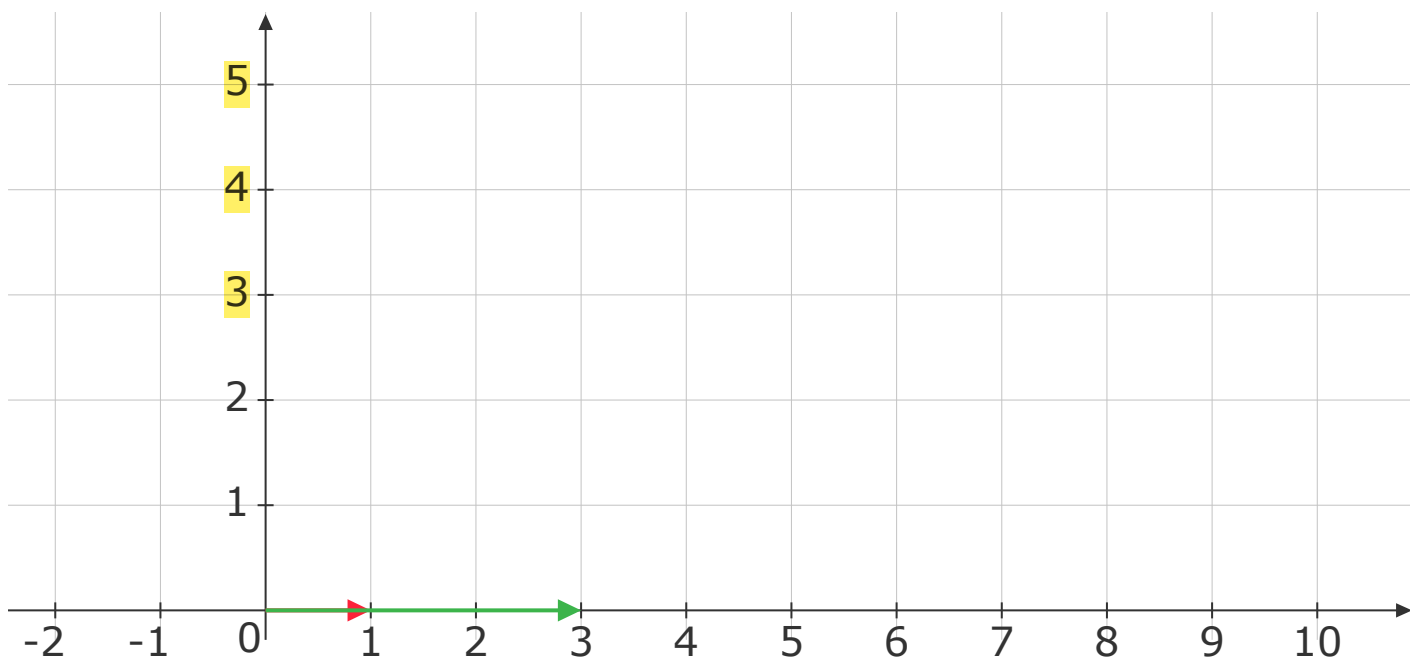
3.1. Case-1

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

(1, 1) and (4, 2) are pointing in different directions.

3.2. Case-2

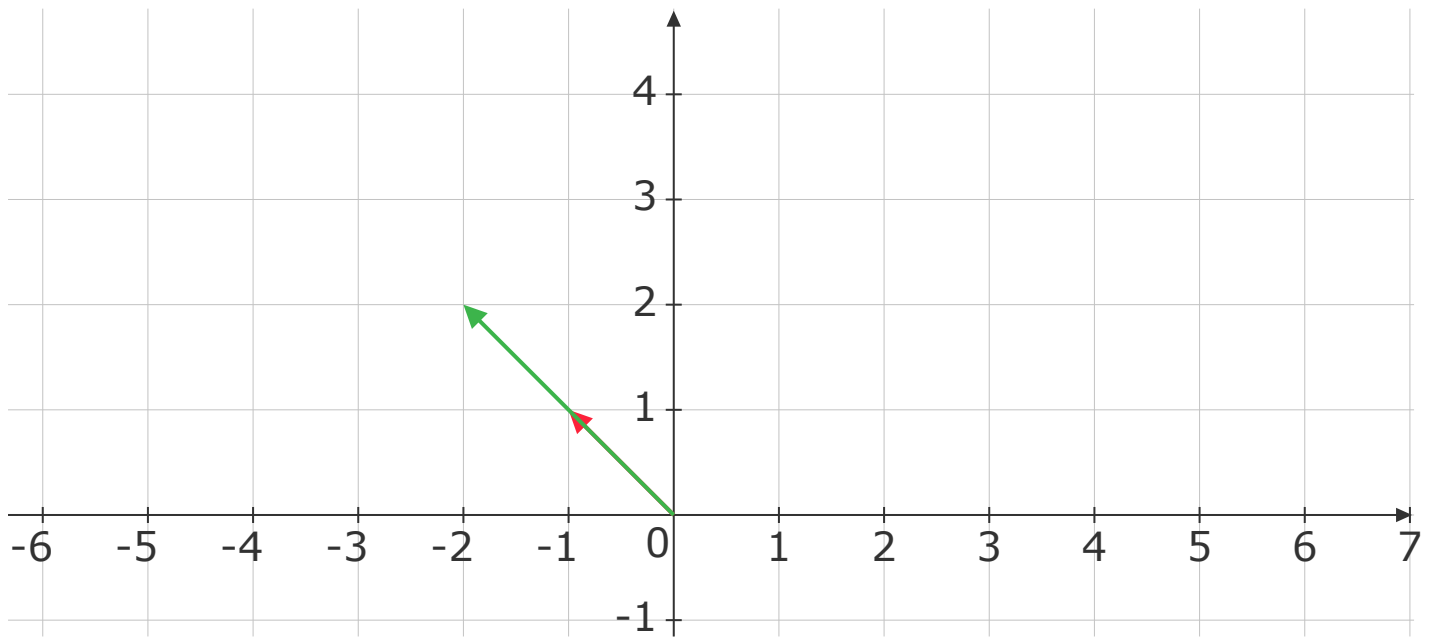
$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$(1, 0)$ and $(3, 0)$ are in the same direction. A doesn't disturb the direction of the input vector.

$$A \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$



$$A \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$(-1, 1)$ and $(-2, 2)$ are in the same direction.

Vectors for which the direction is preserved before and after the transformation are called eigenvectors. In this case, $(-1, 1)$ and $(1, 0)$ are eigenvectors of A . The extent by which the eigenvectors are scaled are called the eigenvalues. 2 and 3 are the corresponding eigenvalues.

Remark: The idea of the direction being preserved is a somewhat crude way of motivating the concept of eigenvalues and eigenvectors. Stick to the proper definition in the next section once you have got a rough idea of the concept.

4. Eigenvalues and eigenvectors: Definition

Formally, for a square matrix A of shape $n \times n$, a non-zero vector $v \in \mathbb{R}^n$ is called an eigenvector with eigenvalue λ if $Av = \lambda v$.

$$Av = \lambda v$$

For example,

- $(1, 0)$ is an eigenvector of $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$ with eigenvalue 3.
- $(-1, 1)$ is an eigenvector of $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$ with eigenvalue 2.

Why should v be non-zero? What would happen if $v = 0$?

$$A0 = \lambda 0 \text{ for all } \lambda \in \mathbb{R}$$

We can't really pin down a particular value of λ and call that the eigenvalue.

What happens when $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$?

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Every vector (v_1, v_2) is an eigenvector with eigenvalue 0.

An eigenvector cannot have two distinct eigenvalues:

$$\begin{aligned} Av &= \lambda_1 v \\ Av &= \lambda_2 v \end{aligned} \implies \lambda_1 = \lambda_2$$

However, a single eigenvalue can be shared by multiple eigenvectors as we will see soon.

5. Eigenspace

If v is an eigenvector of A with eigenvalue λ , is v unique?

No, it is not unique. Let us consider the pair (λ, v) :

$$Av = \lambda v$$

Then:

$$\begin{aligned} A(kv) &= k(Av) \\ &= k(\lambda v) \\ &= \lambda(kv) \end{aligned}$$

Therefore, kv is an eigenvector with eigenvalue λ . In fact, if u and v are eigenvectors with the same eigenvalue λ .

$$\begin{aligned} A(u + v) &= Au + Av \\ &= \lambda u + \lambda v \\ &= \lambda(u + v) \end{aligned}$$

Therefore, $u + v$ is an eigenvector of A with eigenvalue λ .

Eigenspace, $E(\lambda)$

$$E(\lambda) = \{x : Ax = \lambda x, x \in \mathbb{R}^n\}$$

The eigenspace is actually a subspace of $\mathbb{R}^n$. We have shown the first two points in the preceeding discussion.

- closed under addition
- closed under scalar multiplication
- $0 \in E(\lambda)$ because $A0 = \lambda 0$, however, 0 is not an eigenvector

$E(\lambda)$ contains the zero vector and the set of all eigenvectors of A corresponding to λ . Formally,

Eigenspace of an eigenvalue λ , denoted as $E(\lambda)$, is the set of all eigenvectors of A with eigenvalue λ along with the zero vector.

5.1. Example-1



$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

0 is the only eigenvalue here, repeated twice.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$E(0) = \mathbb{R}^2$$

5.2. Example-2

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

1 is an eigenvalue. In fact, 1 is the only eigenvalue of A , repeated twice.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 1 \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$E(1) = \mathbb{R}^2$$

Can you have the following?

$$Au = \lambda_1 u$$

$$Av = \lambda_2 v$$

A matrix could have multiple eigenvalues.

6. Characteristic polynomial

How do we find the set of all eigenvalues for a matrix A ?

All the steps below are reversible.

x is an eigenvector
with eigenvalue λ

$$\begin{aligned} \iff Ax &= \lambda x, & x \neq 0 \\ \iff Ax &= \lambda Ix, & x \neq 0 \\ \iff Ax - \lambda Ix &= 0, & x \neq 0 \\ \iff (A - \lambda I)x &= 0, & x \neq 0 \\ \iff \mathcal{N}(A - \lambda I) &\neq \{0\} \\ \iff A - \lambda I &\text{ is not invertible} \\ \iff |A - \lambda I| &= 0 \end{aligned}$$

x is an eigenvector with eigenvalue λ if and only if x is a solution to this homogeneous equation and $x \neq 0$. Therefore, $A - \lambda I$ has infinitely many solutions. This can happen only if $A - \lambda I$ is non-invertible, which in turn means that $|A - \lambda I| = 0$.

λ is an eigenvalue of A if and only if $|A - \lambda I| = 0$. Note that the eigenvectors do not make an appearance here.

We can temporarily forget about the eigenvectors and just worry about computing the eigenvalues.

$|A - \lambda I|$ is called the characteristic polynomial. This is a polynomial with degree n . The roots of the characteristic polynomial are the eigenvalues. Therefore, an $n \times n$ matrix can have at most n eigenvalues and also, the eigenvalues could repeat.

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\begin{vmatrix} 3-\lambda & 1 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)(2-\lambda) = 0$$

$$\implies \lambda = 2, 3$$

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{vmatrix} 1-\lambda & 3 & 3 \\ 3 & 1-\lambda & -1 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda) \cdot [(1-\lambda)^2 - 9] = 0$$

$$(2-\lambda)(4-\lambda)(-2-\lambda) = 0$$

$$\lambda = 2, 4, -2$$

Lemma: Eigenvectors corresponding to distinct eigenvalues of a symmetric matrix are orthogonal.

Proof: Assume that $\lambda_1 \neq \lambda_2$ are eigenvalues of a symmetric matrix A .

$$(\lambda_1, v_1), (\lambda_2, v_2)$$

Then:

$$Av_1 = \lambda_1 v_1$$

$$Av_2 = \lambda_2 v_2$$

Now:

$$v_2^T Av_1 = \lambda_1 v_2^T v_1$$

$$v_2^T A^T v_1 = \lambda_1 v_2^T v_1$$

$$(Av_2)^T v_1 = \lambda_1 v_2^T v_1$$

$$\lambda_2 v_2^T v_1 = \lambda_1 v_2^T v_1$$

$$(\lambda_2 - \lambda_1) v_2^T v_1 = 0$$

Therefore, $v_2^T v_1 = 0$.

7. Characteristic Polynomials: Some Observations

- The characteristic polynomial has degree n .
- There are at most n eigenvalues.
- The sum of the eigenvalues is equal to the trace of the matrix.
- The product of the eigenvalues is equal to the determinant of the matrix.

$$|A - \lambda I| = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$$

If you set $\lambda = 0$ then $|A| = \lambda_1 \cdots \lambda_n$.

- There are matrices that have no eigenvalues.

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 1 - \lambda & -1 \\ 1 & 1 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)^2 = -1$$

A doesn't have any real eigenvalues.

Remark: For weeks 3 and 4, we are only concerned with real vector spaces.

Can a matrix have eigenvectors without having any eigenvalues?

No. This is absurd. Eigenvectors and eigenvalues are inseparable.

8. Computing Eigenvectors (Eigenspace)

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

A has eigenvalues 3, 2. Let us now compute the eigenvectors for $\lambda = 2$.

$$(A - 2I)v = 0$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

v_2 is a free variable. Setting $v_2 = 1$, $v_1 = -1$. Therefore, $(-1, 1)$ is an eigenvector. In fact, the eigenspace $E(2)$:

$$E(2) = \text{span}\{(-1, 1)\}$$

You have to essentially solve the system of homogeneous equations.

Alternatively, you can look at it as first finding a basis for the nullspace of

$$A - \lambda I.$$

$$(A - 3I)v = 0$$

$$\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



$$\begin{bmatrix} 0 & \mathbf{1} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The second column contains the pivot. So the second variable is the pivot variable. v_1 is the free variable. So setting $v_1 = 1$, we get $v_2 = 0$. Therefore, $(1, 0)$ is an eigenvector.

$$E(3) = \text{span}\{(1, 0)\}$$

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

We know that $\lambda = 2, 4, -2$.

$$(A + 2I)v = 0$$

$$\begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & -1 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

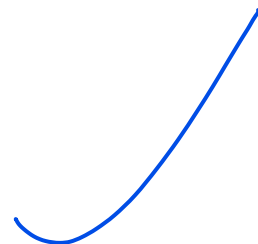
$$\begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & -1 \\ 0 & 0 & 4 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 3 & 3 & 3 \\ 0 & 0 & -4 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\begin{aligned}
 & \begin{bmatrix} 3 & 3 & 3 \\ 0 & 0 & -4 \\ 0 & 0 & 4 \end{bmatrix} \xrightarrow{R_1 \rightarrow \frac{R_1}{3}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & 4 \end{bmatrix} \\
 & \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & 4 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & 0 \end{bmatrix} \\
 & \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow \frac{R_2}{-4}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \\
 & \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

$$\begin{bmatrix} \mathbf{1} & 1 & 0 \\ 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- v_1 and v_3 are pivot (dependent) variables
- v_2 is the only free variable

$$\begin{aligned}
 v_1 &= -1 \\
 v_2 &= 1 \\
 v_3 &= 0
 \end{aligned}$$



Therefore, $(-1, 1, 0)$.

$$E(-2) = \text{span}\{(-1, 1, 0)\}$$

So far all the examples that we have looked have an eigenspace that is one-dimensional. Is this always the case?

No. For example, if $A = I$, the eigenspace is $\mathbb{R}^n$.

9. Eigenthings of Special Matrices

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

The characteristic polynomial of D is:

$$|D - \lambda I| = 0$$

$$\begin{vmatrix} 1 - \lambda & 0 & 0 \\ 0 & 2 - \lambda & 0 \\ 0 & 0 & 3 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(2 - \lambda)(3 - \lambda) = 0$$

The eigenvalues of a diagonal matrix are its diagonal elements. One can directly read the eigenvalues off the diagonal matrix with just a lookup. $D[i][i]$. This is $O(1)$, computationally efficient.

What are the eigenvectors of D ? The eigenvectors are the elements of the standard basis.

$$D = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 0 \end{bmatrix}$$

$E(1)$ is a two-dimensional subspace of $\mathbb{R}^3$.

$$T = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

T is upper-triangular.

$$|T - \lambda I| = 0$$

$$\begin{vmatrix} 1 - \lambda & 3 & 1 \\ 0 & 2 - \lambda & 2 \\ 0 & 0 & 3 - \lambda \end{vmatrix} = 0$$

$$(3 - \lambda)(1 - \lambda)(2 - \lambda) = 0$$

The eigenvalues of an upper-triangular (or lower triangular) matrix are the diagonal elements.

Projection matrices:

$$P^2 = P \quad \text{and} \quad P^T = P$$

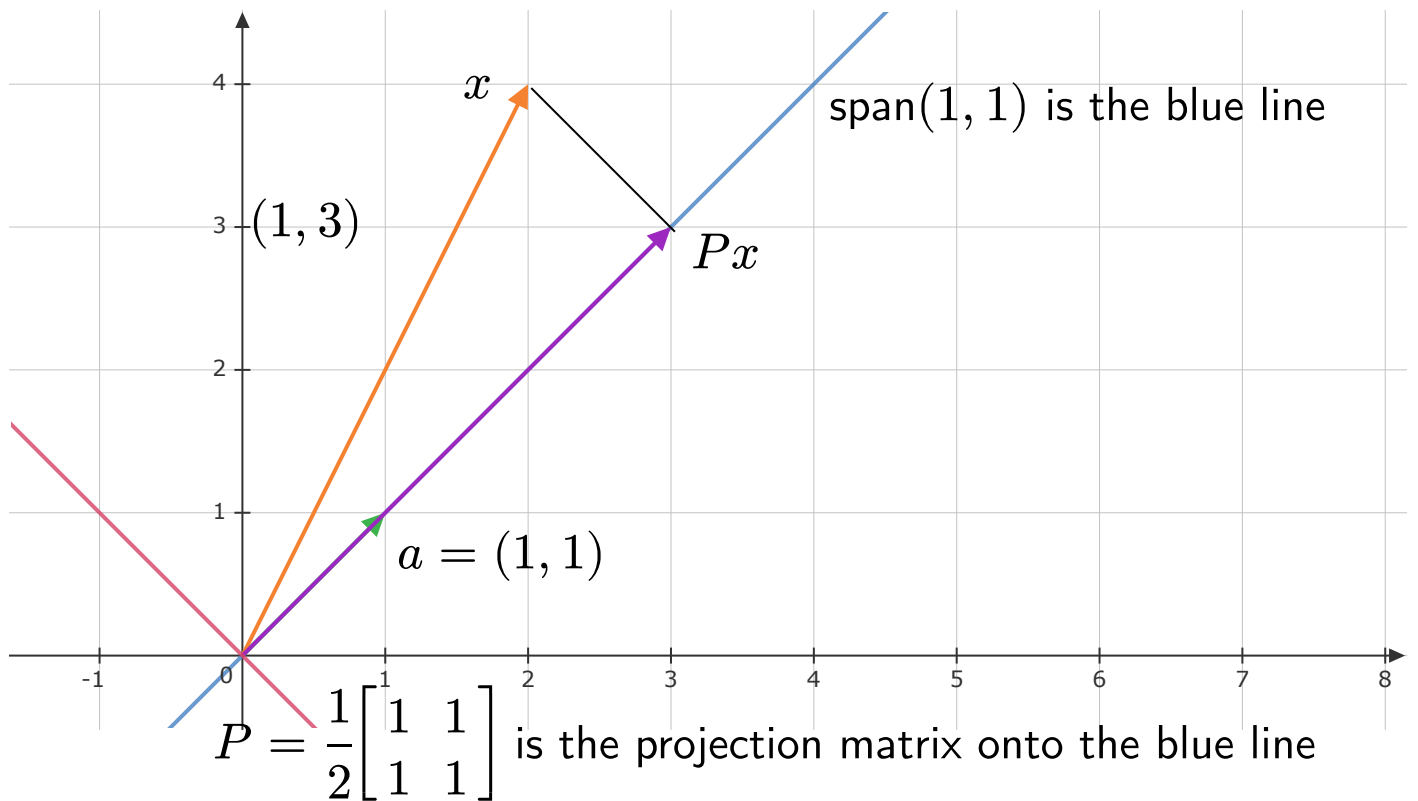
The eigenvalues are 0 and 1. If you pick an element v from the column space of P , what is $Pv = v$. 1 is an eigenvalue and every element in the column space of P is an eigenvector.

$$aa^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Outer products characterize rank-1 matrices. Any matrix can be decomposed as a sum of rank-1 matrices (SVD). This can help you break down a matrix as a sum of smaller matrices.

- 50×5 matrix
- 50 entries in an excel sheet with 5 features describing each entry

Low rank approximation. LoRA in LLMs.



$$a^T a = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2$$

$$P = \frac{aa^T}{a^T a} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{aligned} P^2 &= \frac{1}{4} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \\ &= P \end{aligned}$$



Verify. If $x = (2, 4)$ then Px is the projection of x onto the line passing through $(1, 1)$.

$$P \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$P \begin{bmatrix} k \\ k \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} k \\ k \end{bmatrix} = 1 \cdot \begin{bmatrix} k \\ k \end{bmatrix}$$

Notice that (k, k) is an eigenvector of P with eigenvalue 1.

$$P \begin{bmatrix} k \\ -k \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} k \\ -k \end{bmatrix} = 0 \cdot \begin{bmatrix} k \\ -k \end{bmatrix}$$

Notice that $(k, -k)$ is an eigenvector of P with eigenvalue 0.

